使用 AdamW 训练的资源开销计算

B = batch_size, H = head_nums, T = context_length, V = vocab_size

Parameters: $\text{vocab_size} \times d_model + (2 \times d_model + d_model \times d_model \times 4 + d_ff \times d_model \times 3) \times \text{num_layers} + d_model + d_model \times \text{vocab_size}$

Activations: $(2 \times B \times T \times d_model$

MHA: $+ \underbrace{3 \times d_model \times B \times T}_{\text{QKV}} + \underbrace{B \times H \times T \times T \times 2}_{QK^T,\ \text{softmax}} + \underbrace{d_model \times B \times T}_{\text{weighted_sum}}$

$+ \underbrace{d_model \times B \times T}_{O} +$

FFN: $\underbrace{d_ff \times B \times T \times 2 + d_ff \times B \times T}_{W_1, W_2,\ \text{silu}} + \underbrace{d_ff \times B \times T}_{\text{element-wise product}} + \underbrace{d_model \times B \times T}_{W_3}$

子层的输出，保留下来backward时求梯度要用

$) \times \text{num_layers} + \underbrace{d_model \times B \times T}_{\text{final_RMSNorm}} + \underbrace{\text{vocab_size} \times B \times T \times 2}_{\text{out_embedding + cross_entropy on logits}}$

中间有 softmax

Gradients = Parameters

Optimizer state = 2 × Gradients